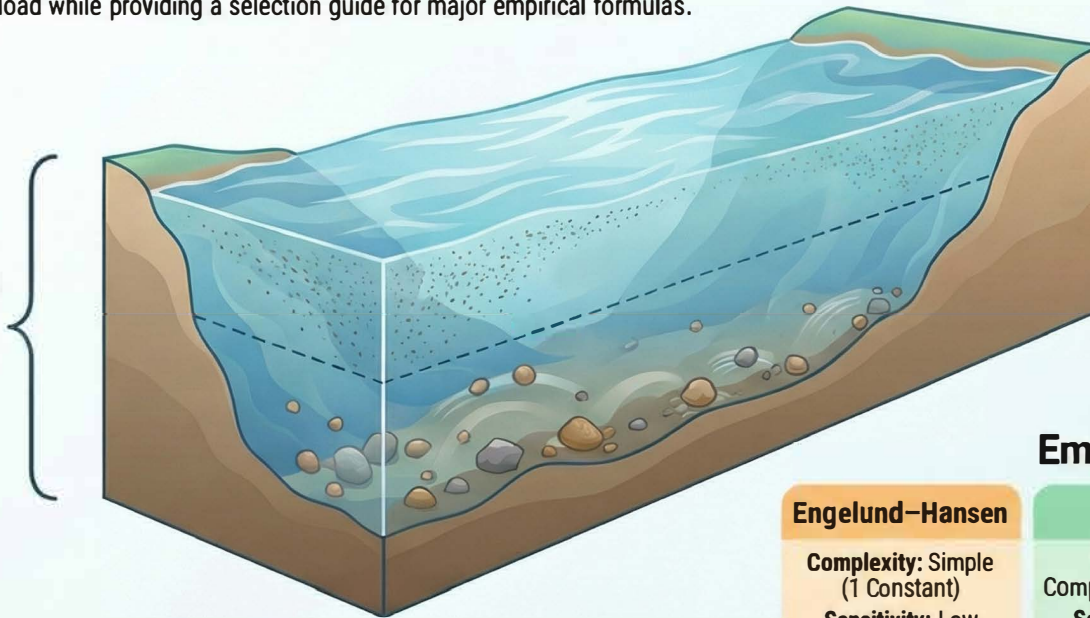


Total Sediment Load: Quantifying River Transport

This infographic integrates sediment transport fundamentals (bed forms, bed load, and suspended load) into a unified Total Load framework. It distinguishes between hydraulic-controlled bed material and supply-controlled wash load while providing a selection guide for major empirical formulas.

The Total Load Equation:

$$q_s = q_b + q_{ss}$$



Bed-Material Load vs. Wash Load



Wash Load:
Supply-controlled
fine particles



Suspended
($P < 1.2$)

Rouse Number (P)
Defines Transport Mode



Mixed
($1.2 < P < 2.5$)



Bed-Material Load:
Derived from the bed



Bed Load
($P > 2.5$)

Empirical Formula Selection

Engelund-Hansen

Complexity: Simple
(1 Constant)
Sensitivity: Low
Best Use Case:
Quick estimates;
Sand-bed rivers

Yang

Complexity:
Computed Coefficients
Sensitivity: High
Best Use Case:
Fine sand/silt;
Physically insightful

Molinas-Wu

Complexity:
Fixed Constants
Sensitivity: Moderate
Best Use Case:
Balanced and stable
numerical modeling

Ackers-White

Complexity:
Multi-parameter
Sensitivity: High
Best Use Case:
Sand-gravel mixtures;
Detailed engineering

Engineering Application & Selection

The “2–5×” Variation Rule:
Using the wrong formula
can easily lead to a 2–5×
difference in results.



**Selection Based on
Sediment Size**



Never Mix Shear Stress Types:
Do not use grain shear stress with
formulas calibrated for total shear stress.

Recommended Method Based on Available Data

Only Depth, Slope, Size

Engelund-Hansen

Velocity (U) Known

Yang or Molinas-Wu

Grain Shear Stress Known

Einstein Divided Approach